Yi (Sean) Xia

Applied AI & ML Software Engineer

Open to: Applied AI Engineer; Machine Learning Engineer; Computer Vision Engineer; Data Scientist / Applied Scientist; AI Research Engineer; Software Engineer (AI/ML)

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SUMMARY

Applied AI engineer and PhD candidate who ships production AI systems across media and public-sector digital forensics. Four years at TOHU MEDIA delivering object detection, OCR and action-recognition pipelines, plus an AUT Research Assistant project for Tonga Police: an offline LLM application for analysing reports from the Autopsy digital-forensics platform and assessing risk. Hands-on across Python, PyTorch, Docker, local/cloud inference, end-user workflows and privacy-sensitive deployment. Two first-author Q1 journal papers, eight peer-reviewed papers and 70+ citations.

TECHNICAL SKILLS

Programming and ML: Python, Java, PyTorch, TensorFlow/Keras, scikit-learn, NumPy, pandas, Hugging Face.

Computer vision: YOLO family, object detection, object tracking, OCR, video analytics, action recognition, segmentation, classification.

Deployment: Docker, containerised inference, Windows/Linux, Git/GitHub, CUDA, cloud GPU servers, standalone applications.

Research methods: GCNs, CNNs, attention/Transformers, skeleton-based HAR, frequency-domain learning.

LLM systems: Ollama/Gemma, Claude/OpenAI APIs, multi-provider integration, offline inference, prompt engineering, rule-plus-LLM pipelines.

EXPERIENCE

Artificial Intelligence Developer — Feb 2022 - May 2026

TOHU MEDIA, Auckland - full-time June 2025 to May 2026; part-time alongside master's and PhD study before that

- Owned the company's AI capability end-to-end: dataset scope, PyTorch training/evaluation, Docker-packaged inference and deployment to company cloud GPU servers.
- Designed and shipped three production AI systems - object detection, OCR and action recognition - replacing manual transcription work on client video footage (specific deliverables under NDA).
- Engineered for variable cinematography, dense overlay text, low-contrast archives, long-form takes and workloads beyond single-workstation GPU limits.
- Worked with editors and the creative director on architecture, useful outputs and latency/accuracy trade-offs while preserving client data sovereignty.

Research Assistant - LLM Digital Forensics — Oct 2025 - Feb 2026

Auckland University of Technology; client: Tonga Police

- Built and delivered Autopsy AI, a Windows application that parses Autopsy HTML case reports, combines rule-based risk scoring with LLM analysis, and generates investigator-ready reports.
- Designed a 100% offline deployment with Ollama/Gemma, English and Tongan risk keywords, multi-provider LLM support and a standalone executable, keeping sensitive forensic data on-device.

PhD Researcher - Computer Vision and Human Activity Recognition — Oct 2023 - present

Institute of Robotics and Vision, Auckland University of Technology

- Designed FATL-GCN for skeleton-based human activity recognition; published in Pattern Recognition (Q1, IF 7.6) in 2026.
- First-authored a generalisation survey in Computer Science Review (Q1 review, IF 17.5, 2026) and model papers at ICONIP 2024 and ICONIP 2026 (accepted); owned model design, GPU experiments, writing and rebuttal.

Research Assistant - Computer Vision **Nov 2022 - Jul 2023**

School of Engineering, Computer and Mathematical Sciences, AUT

- Co-built KiwiDetector (improved YOLOv7) and KiwiTracker for real-time orchard video; lead paper has 50+ citations.
- Co-authored three peer-reviewed papers across IVCNZ 2022, SAI Intelligent Systems 2023 and PSIVT 2023.

Teaching Assistant **Mar 2025 - present**

Auckland University of Technology

- TA on computer-science courses including programming, AI and computer vision; deliver classes and tutorials, prepare teaching materials and mark assessment.

Tech Service **Apr - Dec 2023**

PB Tech, Auckland

- Handled hardware diagnostics, software troubleshooting, customer returns and custom PC builds; strengthened plain-English technical communication with non-specialist users.

Junior Lecturer / Co-Founder **2015 - 2021**

Baiyin Vocational and Technology College / Baiyin Pinuo Training School, China

- Taught Computer Fundamentals, Java, IoT applications and RFID technologies to about 120 students per year for four academic years.
- Co-founded a private training school; built curriculum, ran academic operations and finance, then divested before relocating to New Zealand.

EDUCATION

PhD, Computer and Information Sciences **Oct 2023 - present**

Auckland University of Technology

AUT PhD Full Scholarship. Thesis on skeleton-based human activity recognition submitted; oral examination February 2027.

Master of Computer and Information Sciences **2021 - 2023**

Auckland University of Technology

Specialisation: Computer Vision.

BEng, Logistics Engineering (Software) **2011 - 2015**

Dalian Neusoft University of Information

SELECTED PUBLICATIONS

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Beyond Benchmark Accuracy: A Generalization-Centered Analysis of Deep Learning for Skeleton-Based Human Activity Recognition." Computer Science Review, vol. 62. Q1 review journal, IF = 17.5. doi.org/10.1016/j.cosrev.2026.101030

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition." Pattern Recognition. Q1 journal, IF 7.6. doi.org/10.1016/j.patcog.2026.113146

Y. Xia, S. Yongchareon, R. Lutui (2026). "ESA-GCN: Efficient Semantic-Aware Graph Convolution for Skeleton-Based Action Recognition." ICONIP 2026, Springer LNCS, accepted, to appear.

Y. Xia, M. Nguyen, W. Q. Yan (2022). "A Real-Time Kiwifruit Detection Based on Improved YOLOv7." IVCNZ 2022, Springer LNCS. 50+ citations. doi.org/10.1007/978-3-031-25825-1_4

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2025). "Dynamic Self-Attention Gated Spatial-Temporal GCN for Skeleton-Based HAR." ICONIP 2024, Springer LNCS. doi.org/10.1007/978-981-96-6588-4_14. Full publication list on Google Scholar.

AWARDS, SERVICE AND COMMUNICATION

- AUT PhD Full Scholarship - tuition + living stipend, Oct 2023 - Oct 2026.
- SECMS Research Assistant Fund and AUT Summer Research Assistant Scholarship.
- Peer reviewer for ACM TOMM, International Agrophysics and IEEE PIMRC.
- Tertiary teaching in China and New Zealand: four years as a lecturer in China, and seven AUT courses since 2025; bilingual Mandarin Chinese and professional English.